3. *The GraphStage API*—This is the lowest-level API, which the Akka Streams library uses to create GraphDSL and the Flow API. It is very unlikely that you will ever need such a low-level composition.

In this chapter, you learn about the Flow API, which you used in chapter 10. It has three types of stages: the `Source` is the producer, a `Sink` is the consumer, and, in the middle, the `Flow` is where the transformations take place.

**NOTE** The source code for this chapter is available at www.manning.com/books/akka-in-action-second-edition or https://github.com/franciscolopez sancho/akka-topics/tree/main/chapter16.

# 16.1 Processing elements through services

When you call external services, you may find yourself in either of two situations:

- You call a service that is local and CPU-bounded, which means its execution consumes the resources of the computer.
- You call an external service whose computation is independent of your running application: that is, the resources consumed by this service are independent of those of the running stream.

## 16.1.1 CPU-bounded services